

Mathematics Section A

1. Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x+[\frac{x}{3}]}{3}\right)$ is $[\alpha, \beta]$, then $\alpha^2 + \beta^2$ is equal to:
(A) 2 (B) 5 (C) 10 (D) 13
2. Let one root of the quadratic equation in x : $(k^2 - 15k + 27)x^2 + 9(k-1)x + 18 = 0$ be twice the other. Then the length of the latus rectum of the parabola $y^2 = 6kx$ is equal to:
(A) 4 (B) 6 (C) 8 (D) 12
3. Let e_1 and e_2 be two distinct roots of the equation $x^2 - ax + 2 = 0$. Let the sets $\{a \in \mathbb{R} : e_1 \text{ and } e_2 \text{ are the eccentricities of hyperbolas}\} = (\alpha, \beta)$, and $\{a \in \mathbb{R} : e_1 \text{ and } e_2 \text{ are the eccentricities of an ellipse and a hyperbola, respectively}\} = (\gamma, \infty)$. Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:
(A) 18 (B) 22 (C) 26 (D) 34
4. Let the set of all values of $k \in \mathbb{R}$ such that the equation $z(\bar{z} + 2 + i) + k(2 + 3i) = 0, z \in \mathbb{C}$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha + \beta)$ is equal to:
(A) -10 (B) -8
(C) $10\sqrt{13}$ (D) $8\sqrt{13}$
5. The value of $1^3 - 2^3 + 3^3 - \dots + 15^3$ is:
(A) 1706 (B) 1856 (C) 1982 (D) 2403
6. The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:
(A) $\frac{34}{9}$ (B) $\frac{34}{13}$
(C) $\frac{32}{9}$ (D) $\frac{32}{13}$
7. The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:
(A) 2670 (B) 2840 (C) 2920 (D) 3600
8. If the coefficients of the middle terms in the binomial expansions of $(1 + ax)^{26}$ and $(1 - ax)^{28}, a \neq 0$, are equal, then the value of α is:
(A) 1 (B) $\frac{14}{13}$
(C) $\frac{27}{7}$ (D) $\frac{7}{27}$
9. A data consists of 20 observations $x_1, x_2, \dots, x_{20}$. $\sum_{i=1}^{20} (x_i + 5)^2 = 2500$ and $\sum_{i=1}^{20} (x_i - 5)^2 = 100$, then the ratio of mean to standard deviation of this data is:
(A) 2:1 (B) 3:1 (C) 3:2 (D) 4:1
10. A bag contains $(N + 1)$ coins - N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:
(A) 5 (B) 7 (C) 8 (D) 9
11. If the eccentricity e of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, passing through $(6, 4\sqrt{3})$, satisfies $15(e^2 + 1) = 34e$, then the length of the latus rectum of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2 + 1} = 1$ is:
(A) 10 (B) 20 (C) 25 (D) 30
12. Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2 = 12x$ such that the ordinates of points P and Q are in the ratio 1:2. If the chord PQ subtends an angle α at the focus of the parabola, then $\sin \alpha$ is equal to:
(A) $\frac{3}{5}$ (B) $\frac{4}{5}$
(C) $\frac{5}{13}$ (D) $\frac{12}{13}$
13. Let $0 < \alpha < 1, \beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1 - \alpha) + \tan^{-1}(1 - \beta) = \frac{\pi}{4}$. Then $6(\alpha + \beta)$ is equal to:
(A) 6 (B) 7 (C) 8 (D) 9
14. Let $S = \{\theta \in (-2\pi, 2\pi) : \cos \theta + 1 = \sqrt{3} \sin \theta\}$. Then $\sum_{\theta \in S} \theta$ is equal to:
(A) $-\frac{2\pi}{3}$ (B) $-\frac{4\pi}{3}$
(C) $\frac{2\pi}{3}$ (D) $\frac{4\pi}{3}$
15. Let the image of the point $P(1, 6, a)$ in the line $L : \frac{x}{1} = \frac{y-1}{2} = \frac{z-a+1}{b}, b > 0$, be $(\frac{a}{3}, 0, a + c)$. If $S(\alpha, \beta, \gamma), \alpha > 0$ is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha\beta + \gamma$ is equal to:
(A) 19 (B) 20 (C) 21 (D) 22
16. Let a line L be perpendicular to both the lines $L_1 : \frac{x+3}{5} = \frac{z+5}{7}$ and $L_2 : \frac{x-2}{1} = \frac{y-4}{4} = \frac{z-6}{7}$. If θ is the acute angle between the lines L and L_2 , $\frac{x-\frac{5}{2}}{2} = \frac{y-\frac{4}{1}}{1} = \frac{z}{2}$, then $\tan \theta$ is equal to:
(A) $\frac{3}{2}\sqrt{2}$ (B) $\frac{5}{2}\sqrt{2}$
(C) $\frac{5}{3}\sqrt{2}$ (D) $\frac{4}{3}\sqrt{2}$
17. The value of $\lim_{x \rightarrow 0} \left(\frac{x^2 \sin^2 x}{x^2 - \sin^2 x} \right)$ is:

- (A) 2 (B) 3 (C) 4 (D) 6
18. The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{32 \cos^4 x}{1 + e^{\sin x}} \right) dx$ is:
 (A) $4\pi + 2$ (B) $3\pi + 8$ (C) $3\pi + 4$ (D) $4\pi + 3$
19. The area of the region $\{(x, y) : 0 \leq y \leq 6 - x, y^2 \geq 4x - 3, x \geq 0\}$ is:
 (A) 8 (B) 9 (C) 12 (D) 15
20. Let e be the base of natural logarithm and let $f : \{1, 2, 3, 4\} \rightarrow \{1, e, e^2, e^3\}$ and $g : \{1, e, e^2, e^3\} \rightarrow \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x) = [f^{-1}\{g^{-1}(\frac{1}{2})\}]^x$, then the area of the region $R = \{(x, y) : x^2 \leq y \leq \phi(x), 0 \leq x \leq 1\}$ is:
 (A) $\frac{3 - \log_e(2)}{3 \log_e(2)}$ (B) $\frac{1}{3 \log_e(2)}$
 (C) $3 + \log_e(2)$ (D) $\frac{3 + \log_e(2)}{2 + \log_e(3)}$

Mathematics Section B

21. Let $A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ satisfy $A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ for some $\alpha, \beta \in \mathbb{R}$. Then $(\alpha - \beta)^2$ is equal to ____.
22. Let the centre of the circle $x^2 + y^2 + 2gx + 2fy + 25 = 0$ be in the first quadrant and lie on the line $2x - y = 4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x = 1$ is ____.
23. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{c} \cdot (\vec{a} - 2\vec{b})$ equal to ____.
24. For the functions $f(\theta) = \alpha \tan^2 \theta + \beta \cot^2 \theta$, and $g(\theta) = \alpha \sin^2 \theta + \beta \cos^2 \theta$, $\alpha > \beta > 0$, let $\min_{0 < \theta < \frac{\pi}{2}} f(\theta) = \frac{m}{n}$ and $\max_{0 < \theta < \pi} g(\theta) = \frac{p}{q}$. If the first term of a G.P. is $(\frac{a}{2b})$, its common ratio is $(\frac{2b}{a})$ and the sum of its first 10 terms is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to ____.
25. Let $y = y(x)$ be the solution of the differential equation $(x^2 - x\sqrt{x^2 - 1}) dy + (y(x - \sqrt{x^2 - 1}) - x) dx = 0$, $x \geq 1$. If $y(1) = 1$, then the greatest integer less than $y(\sqrt{5})$ is ____.

Physics Section A

26. The density ρ of a uniform cylinder is determined by measuring its mass m , length l and diameter d . The measured values of m, l and d are 97.42 ± 0.02 g, 8.35 ± 0.05 mm and 20.20 ± 0.02 mm, respectively. Calculated percentage fractional error in ρ is ____.
 (A) 0.63% (B) 0.82% (C) 0.72% (D) 0.25%
27. The potential energy of a particle changes with distance x from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where A and B are constant with appropriate dimensions. The dimensions of AB are ____.
 (A) $[M^1 L^{5/2} T^{-2}]$
 (B) $[M^{3/2} L^{5/2} T^{-2}]$
 (C) $[M^1 L^2 T^{-2}]$
 (D) $[M^1 L^{7/2} T^{-2}]$
28. The rain drop of mass 1 g, starts with zero velocity from a height of 1 km. It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is ____ (take $g = 10 \text{ m/s}^2$)
 (A) -8.75 (B) -8.35 (C) -9.55 (D) -9.98
29. Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pulley which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ . The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is ____ (take $g = 10 \text{ m/s}^2$)